ANY CONCEPCION GONZALEZ RODRIGUES

Molecular Bioengineer | Neural Electrophysiology, HD-MEA & Cell Culture

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RESEARCH PROFILE

Molecular Bioengineer with hands-on experience in neuronal cell culture, microelectrode array electrophysiology and experimental workflow development. Established and optimized *ex vivo* retinal HD-MEA recordings at CRTD and investigated neuronal activity in olfactory cultures and co-cultures at DZNE. Interested in how electrophysiological measurements and closed-loop experimental systems can be used to study neural plasticity, learning and biological computation, particularly in human brain organoid models.

EDUCATION

MSc in Molecular Bioengineering | University of Porto - ICBAS/FEUP
Sep 2022 - Oct 2024 | Porto, Portugal

Thesis: Establishing CMOS-MEA Recordings in Adult Zebrafish Retina

Supervisor: Prof. Dr. Michael Brand - CRTD, Germany

BSc in Bioengineering | University of Porto - ICBAS/FEUP
Sep 2018 - Jun 2022 | Porto, Portugal

Thesis: Development of Optical Sensors Using Molecularly Imprinted Polymers for Ferritin Detection

Supervisor: Dr. Margarida Piloto - ISEP, Portugal

RESEARCH EXPERIENCE

Master's Thesis Researcher – Neural Electrophysiology
Center for Regenerative Therapies Dresden (CRTD), TU Dresden | Feb 2024 – Aug 2024

- Established and optimized an *ex vivo* CMOS-MEA recording workflow for adult zebrafish retina, integrating tissue preparation, perfusion, visual stimulation and pharmacological experiments.
- Troubleshot tissue attachment, recording stability and experimental conditions to improve the reliability of retinal electrophysiology measurements.
- Configured spike-detection and recording analysis parameters in 3Brain BrainWave, exporting neuronal activity data for statistical analysis and visualization in R and Python.
- Analyzed spontaneous and light-evoked retinal activity, characterizing ON, OFF and ON-OFF responses and investigating changes in neuronal firing patterns across stimulation and pharmacological conditions.

Research Intern – Neural Cell Culture and Co-culture
German Center for Neurodegenerative Diseases (DZNE), Dresden | Sep 2023 – Feb 2024

- Seeded and maintained human and rat olfactory cell cultures in 6-well and 24-well plates, including regular medium changes and culture monitoring.
- Prepared olfactory bulb–olfactory epithelium co-cultures for functional studies.
- Prepared media and experimental solutions, following established culture protocols.
- Performed Axion Biosystems MEA recordings during odorant and neurotransmitter stimulation, analyzing neuronal firing and bursting activity.
- Documented experiments and discussed results within an international neuroscience research team.

Research Intern
BioMark, ISEP, Porto | Feb 2022 – Jul 2022

- Contributed to optical sensor development for ferritin detection using molecularly imprinted polymers and UV–Vis spectroscopy

LABORATORY & TECHNICAL SKILLS

Cell culture and experimental biology: Aseptic culture techniques; olfactory cell cultures and co-cultures; culture maintenance; media and solution preparation; biological sample handling.

Molecular biology and imaging: DNA extraction; conventional PCR; agarose gel electrophoresis and gel imaging; immunofluorescence; confocal microscopy.

Electrophysiology and tissue preparation: MEA surface coating and preparation; MEA and HD-MEA recordings (Axion BioSystems, 3Brain); adult zebrafish retinal dissection; *ex vivo* tissue preparation; perfusion; visual and pharmacological stimulation; experimental troubleshooting.

Data analysis: R and Python for neuronal activity analysis, statistical comparisons and visualization; spike-detection parameter configuration in 3Brain BrainWave; firing-rate, bursting and stimulus-response analysis.

Research practices: Experimental documentation, troubleshooting, scientific figures and presentations, collaboration in international research teams.

PROFESSIONAL EXPERIENCE

Account Management Intern
MaxWell Biosystems, Zurich | May 2025 – Apr 2026

- Supported scientific communication on HD-MEA applications involving neural cultures, organoids and tissue preparations.
- Worked with scientific and commercial teams to organize technical information and support webinars and research-focused communications.

TRAINING AND CERTIFICATIONS

- FELASA-accredited laboratory animal training LAS modules C01 and F09-F11
- Zebrafish handling course
- Introduction to Python and Machine Learning for the Biosciences (i3S)

LANGUAGES

Portuguese — native (C2); English — C1; Spanish — B1; German — A1

RESEARCH INTERESTS

Organoid intelligence, neural plasticity, closed-loop electrophysiology and biological computing.

ACADEMIC REFERENCES

Prof. Dr. Michael Brand — Principal Investigator, CRTD, TU Dresden, Germany
Email: michael.brand@tu-dresden.de | Master's Thesis PI

Paul Röppenack — Research Supervisor, CRTD, TU Dresden, Germany
Email: paul.roeppenack@tu-dresden.de | Master's Thesis Supervisor